PART I : PHYSICS

SECTION 1 (Maximum Marks: 21)

- This section contains **SEVEN** questions
- Each question has **FOUR** options [A], [B], [C] and [D]. **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS
- For each question, marks will be awarded in one of the following categories:

Full Marks	: +3	If only the bubble corresponding to the correct option is darkened
Zero Marks	: 0	If none of the bubbles is darkened
Negative Marks:	-1	In all other cases

Q.1 Consider an expanding sphere of instantaneous radius R whose total mass remains constant. The expansion is such that the *instantaneous* density ρ remains uniform throughout the volume. The rate of fractional change in density $\left(\frac{1}{\rho}\frac{d\rho}{dt}\right)$ is constant. The velocity v of any point on the surface of the expanding sphere is proportional to

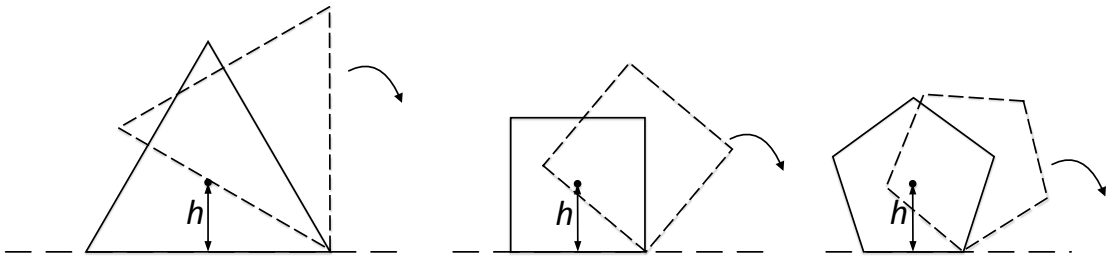
[A] R [B] R^3 [C] $\frac{1}{R}$ [D] $R^{2/3}$

Space for rough work

Answer for the above question

Ans for Q.1: (A)

Q.2 Consider regular polygons with number of sides $n = 3, 4, 5\ldots\ldots$ as shown in the figure. The center of mass of all the polygons is at height h from the ground. They roll on a horizontal surface about the leading vertex without slipping and sliding as depicted. The maximum increase in height of the locus of the center of mass for each polygon is Δ. Then Δ depends on n and h as



[A] $\Delta = h \sin^2\left(\frac{\pi}{n}\right)$

[B] $\Delta = h\left(\frac{1}{\cos\left(\frac{\pi}{n}\right)} - 1\right)$

[C] $\Delta = h \sin\left(\frac{2\pi}{n}\right)$

[D] $\Delta = h \tan^2\left(\frac{\pi}{2n}\right)$

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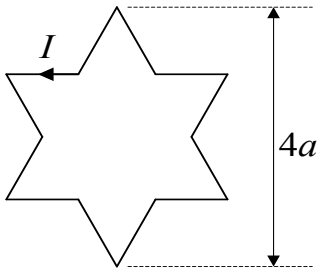
Answer for the above question

Ans for Q.2: (B)

Q.3 A photoelectric material having work-function ϕ_0 is illuminated with light of wavelength $\lambda \left(\lambda < \frac{hc}{\phi_0}\right)$. The fastest photoelectron has a de Broglie wavelength λ_d. A change in wavelength of the incident light by $\Delta\lambda$ results in a change $\Delta\lambda_d$ in λ_d. Then the ratio $\Delta\lambda_d/\Delta\lambda$ is proportional to

[A] λ_d/λ [B] λ_d^2/λ^2 [C] λ_d^3/λ [D] λ_d^3/λ^2

Q.4 A symmetric star shaped conducting wire loop is carrying a steady state current I as shown in the figure. The distance between the diametrically opposite vertices of the star is $4a$. The magnitude of the magnetic field at the center of the loop is



[A] $\frac{\mu_0 I}{4\pi a}\, 6[\sqrt{3}-1]$ [B] $\frac{\mu_0 I}{4\pi a}\, 6[\sqrt{3}+1]$

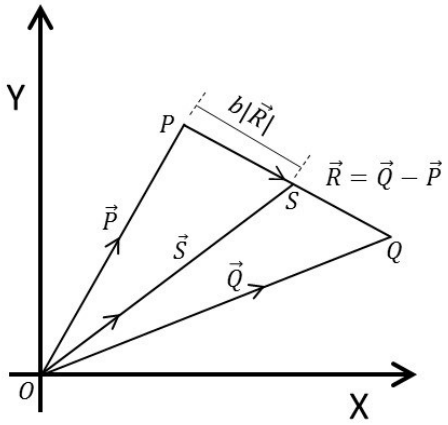
[C] $\frac{\mu_0 I}{4\pi a}\, 3[\sqrt{3}-1]$ [D] $\frac{\mu_0 I}{4\pi a}\, 3[2-\sqrt{3}]$

Space for rough work

Answers for the above questions

Ans for Q.3: (D)	Ans for Q.4: (A)

Q.5 Three vectors $\vec{P}$, $\vec{Q}$ and $\vec{R}$ are shown in the figure. Let S be any point on the vector $\vec{R}$. The distance between the points P and S is $b|\vec{R}|$. The general relation among vectors $\vec{P}$, $\vec{Q}$ and $\vec{S}$ is



[A] $\vec{S} = (1-b)\vec{P} + b\vec{Q}$

[B] $\vec{S} = (b-1)\vec{P} + b\vec{Q}$

[C] $\vec{S} = (1-b^2)\vec{P} + b\vec{Q}$

[D] $\vec{S} = (1-b)\vec{P} + b^2\vec{Q}$

Space for rough work

Answer for the above question

Ans for Q.5: (A)

Q.6 A rocket is launched normal to the surface of the Earth, away from the Sun, along the line joining the Sun and the Earth. The Sun is 3×10^5 times heavier than the Earth and is at a distance 2.5×10^4 times larger than the radius of the Earth. The escape velocity from Earth's gravitational field is $v_e = 11.2$ km s^{-1}. The minimum initial velocity (v_S) required for the rocket to be able to leave the *Sun-Earth system* is closest to
(*Ignore the rotation and revolution of the Earth and the presence of any other planet*)

[A] $v_S = 22$ km s^{-1} [B] $v_S = 42$ km s^{-1}

[C] $v_S = 62$ km s^{-1} [D] $v_S = 72$ km s^{-1}

Q.7 A person measures the depth of a well by measuring the time interval between dropping a stone and receiving the sound of impact with the bottom of the well. The error in his measurement of time is $\delta T = 0.01$ seconds and he measures the depth of the well to be $L = 20$ meters. Take the acceleration due to gravity $g = 10$ ms^{-2} and the velocity of sound is 300 ms^{-1}. Then the fractional error in the measurement, $\delta L/L$, is closest to

[A] 0.2% [B] 1% [C] 3% [D] 5%

Space for rough work

Answers for the above questions

Ans for Q.6: (B)	Ans for Q.7: (B)

SECTION 2 (Maximum Marks: 28)

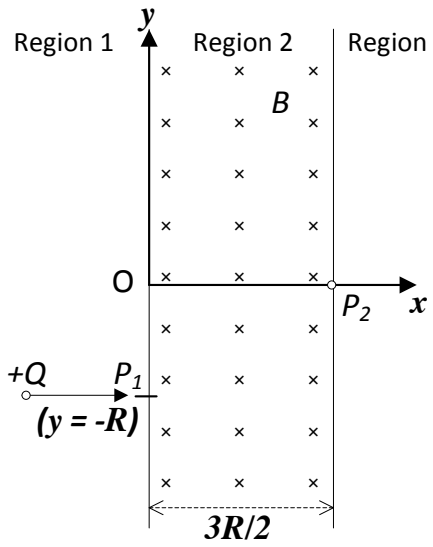
- This section contains **SEVEN** questions
- Each question has **FOUR** options [A], [B], [C] and [D]. **ONE OR MORE THAN ONE** of these four options is(are) correct
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- For each question, marks will be awarded in one of the following categories:

Full Marks	: +4	If only the bubble(s) corresponding to all the correct option(s) is(are) darkened
Partial Marks	: +1	For darkening a bubble corresponding **to each correct option**, provided NO incorrect option is darkened
Zero Marks	: 0	If none of the bubbles is darkened
Negative Marks	: -2	In all other cases

- For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get -2 marks, as a wrong option is also darkened

Space for rough work

Q.8 A uniform magnetic field B exists in the region between $x = 0$ and $x = \frac{3R}{2}$ (region 2 in the figure) pointing normally into the plane of the paper. A particle with charge $+Q$ and momentum p directed along x-axis enters region 2 from region 1 at point P_1 $(y = -R)$. Which of the following option(s) is/are correct?



[A] For $B > \frac{2}{3}\frac{p}{QR}$, the particle will re-enter region 1

[B] For $B = \frac{8}{13}\frac{p}{QR}$, the particle will enter region 3 through the point P_2 on x-axis

[C] When the particle re-enters region 1 through the longest possible path in region 2, the magnitude of the change in its linear momentum between point P_1 and the farthest point from y-axis is $p/\sqrt{2}$

[D] For a fixed B, particles of same charge Q and same velocity v, the distance between the point P_1 and the point of re-entry into region 1 is inversely proportional to the mass of the particle

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Space for rough work

Answer for the above question

Ans for Q.8: (A) and (B)

Q.9 The instantaneous voltages at three terminals marked X, Y and Z are given by

$$V_X = V_0 \sin \omega t,$$
$$V_Y = V_0 \sin\left(\omega t + \frac{2\pi}{3}\right) \text{ and}$$
$$V_Z = V_0 \sin\left(\omega t + \frac{4\pi}{3}\right).$$

An ideal voltmeter is configured to read *rms* value of the potential difference between its terminals. It is connected between points X and Y and then between Y and Z. The reading(s) of the voltmeter will be

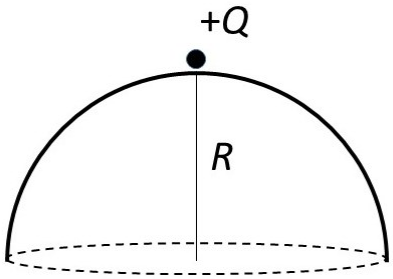
[A] $V_{XY}^{rms} = V_0\sqrt{\frac{3}{2}}$

[B] $V_{YZ}^{rms} = V_0\sqrt{\frac{1}{2}}$

[C] $V_{XY}^{rms} = V_0$

[D] independent of the choice of the two terminals

Q.10 A point charge +Q is placed just outside an imaginary hemispherical surface of radius R as shown in the figure. Which of the following statements is/are correct?



[A] The electric flux passing through the *curved* surface of the hemisphere is $-\frac{Q}{2\varepsilon_0}\left(1-\frac{1}{\sqrt{2}}\right)$

[B] Total flux through the curved and the flat surfaces is $\frac{Q}{\varepsilon_0}$

[C] The component of the electric field normal to the flat surface is constant over the surface

[D] The circumference of the flat surface is an equipotential

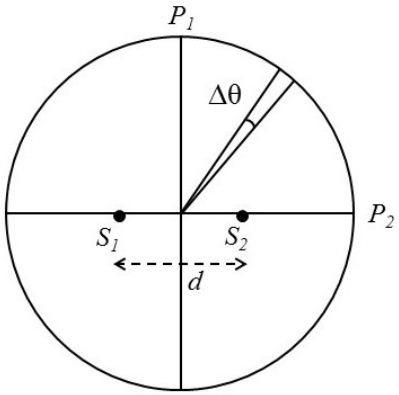
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Space for rough work

Answers for the above questions

Ans for Q.9: (A) and (D)	Ans for Q.10: (A) and (D)

Q.11 Two coherent monochromatic point sources S_1 and S_2 of wavelength $\lambda = 600\,\text{nm}$ are placed symmetrically on either side of the center of the circle as shown. The sources are separated by a distance $d = 1.8\,\text{mm}$. This arrangement produces interference fringes visible as alternate bright and dark spots on the circumference of the circle. The angular separation between two consecutive bright spots is $\Delta\theta$. Which of the following options is/are correct?



[A] A dark spot will be formed at the point P_2

[B] At P_2 the order of the fringe will be maximum

[C] The total number of fringes produced between P_1 and P_2 in the first quadrant is close to 3000

[D] The angular separation between two consecutive bright spots decreases as we move from P_1 to P_2 along the first quadrant

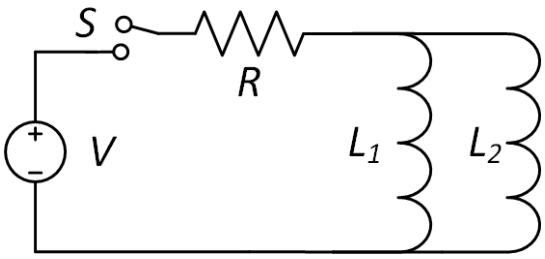
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Space for rough work

Answer for the above question

Ans for Q.11: (B) and (C)

Q.12 A source of constant voltage V is connected to a resistance R and two ideal inductors L_1 and L_2 through a switch S as shown. There is no mutual inductance between the two inductors. The switch S is initially open. At $t = 0$, the switch is closed and current begins to flow. Which of the following options is/are correct?



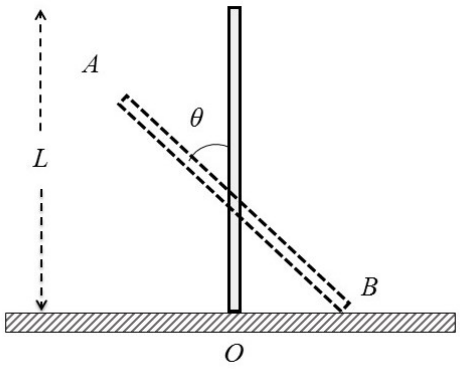
[A] After a long time, the current through L_1 will be $\frac{V}{R}\frac{L_2}{L_1+L_2}$

[B] After a long time, the current through L_2 will be $\frac{V}{R}\frac{L_1}{L_1+L_2}$

[C] The ratio of the currents through L_1 and L_2 is fixed at all times ($t > 0$)

[D] At $t = 0$, the current through the resistance R is $\frac{V}{R}$

Q.13 A rigid uniform bar AB of length L is slipping from its vertical position on a frictionless floor (as shown in the figure). At some instant of time, the angle made by the bar with the vertical is θ. Which of the following statements about its motion is/are correct?



[A] The midpoint of the bar will fall vertically downward

[B] The trajectory of the point A is a parabola

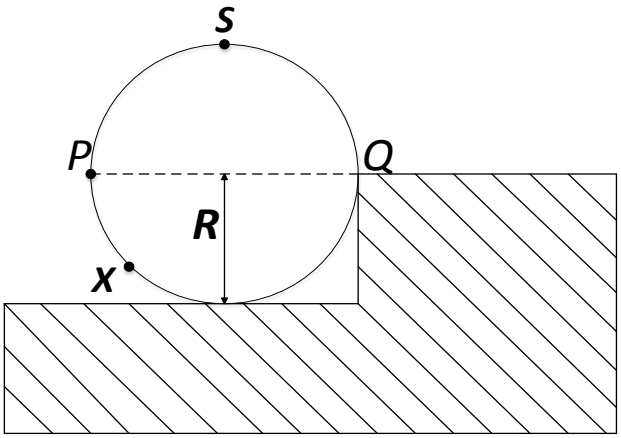
[C] Instantaneous torque about the point in contact with the floor is proportional to $\sin\theta$

[D] When the bar makes an angle θ with the vertical, the displacement of its midpoint from the initial position is proportional to $(1 - \cos\theta)$

Answers for the above questions

Ans for Q.12: (A), (B), and (C)	Ans for Q.13: (A), (C), and (D)

Q.14 A wheel of radius R and mass M is placed at the bottom of a fixed step of height R as shown in the figure. A constant force is continuously applied on the surface of the wheel so that it just climbs the step without slipping. Consider the torque τ about an axis normal to the plane of the paper passing through the point Q. Which of the following options is/are correct?



[A] If the force is applied at point P tangentially then τ decreases continuously as the wheel climbs

[B] If the force is applied normal to the circumference at point X then τ is constant

[C] If the force is applied normal to the circumference at point P then τ is zero

[D] If the force is applied tangentially at point S then $\tau \neq 0$ but the wheel never climbs the step

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Space for rough work

Answer for the above question

Ans for Q.14: {(C)} or {(C) and (D)}

SECTION 3 (Maximum Marks: 12)

- This section contains **TWO** paragraphs
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options [A], [B], [C], and [D]. **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS
- For each question, marks will be awarded in one of the following categories:

Full Marks	: +3	If only the bubble corresponding to the correct option is darkened
Zero Marks	: 0	In all other cases

Space for rough work

PARAGRAPH 1

Consider a simple RC circuit as shown in Figure 1.

Process 1: In the circuit the switch S is closed at $t = 0$ and the capacitor is fully charged to voltage V_0 (*i.e.*, charging continues for time $T >> RC$). In the process some dissipation (E_D) occurs across the resistance R. The amount of energy finally stored in the fully charged capacitor is E_C.

Process 2: In a different process the voltage is first set to $\frac{V_0}{3}$ and maintained for a charging time $T >> RC$. Then the voltage is raised to $\frac{2V_0}{3}$ without discharging the capacitor and again maintained for a time $T >> RC$. The process is repeated one more time by raising the voltage to V_0 and the capacitor is charged to the same final voltage V_0 as in Process 1.

These two processes are depicted in Figure 2.

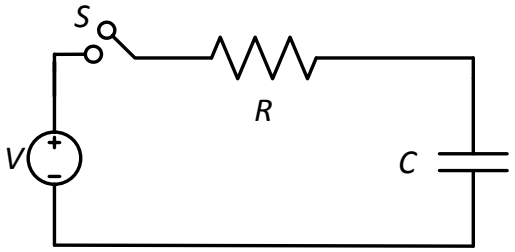


Figure 1

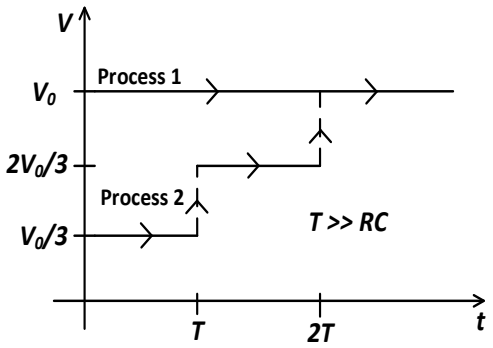


Figure 2

Q.15 In Process 1, the energy stored in the capacitor E_C and heat dissipated across resistance E_D are related by:

[A] $E_C = E_D$ [B] $E_C = E_D \ln 2$

[C] $E_C = \frac{1}{2}E_D$ [D] $E_C = 2E_D$

Q.16 In Process 2, total energy dissipated across the resistance E_D is:

[A] $E_D = \frac{1}{2}CV_0^2$ [B] $E_D = 3\left(\frac{1}{2}CV_0^2\right)$

[C] $E_D = \frac{1}{3}\left(\frac{1}{2}CV_0^2\right)$ [D] $E_D = 3\,CV_0^2$

Answers for the above questions

Ans for Q.15: (A)	Ans for Q.16: (C)

PARAGRAPH 2

One twirls a circular ring (of mass M and radius R) near the tip of one's finger as shown in Figure 1. In the process the finger never loses contact with the inner rim of the ring. The finger traces out the surface of a cone, shown by the dotted line. The radius of the path traced out by the point where the ring and the finger is in contact is r. The finger rotates with an angular velocity ω_0. The rotating ring *rolls without slipping* on the outside of a smaller circle described by the point where the ring and the finger is in contact (Figure 2). The coefficient of friction between the ring and the finger is μ and the acceleration due to gravity is g.

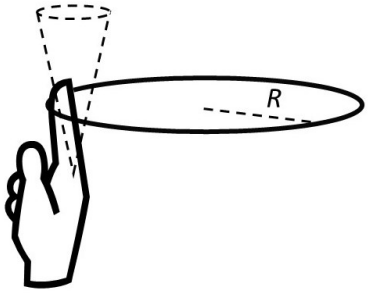


Figure 1

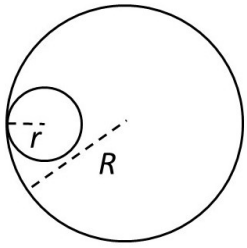


Figure 2

Q.17 The total kinetic energy of the ring is

[A] $M\omega_0^2R^2$ [B] $\frac{1}{2}M\omega_0^2(R-r)^2$

[C] $M\omega_0^2(R-r)^2$ [D] $\frac{3}{2}M\omega_0^2(R-r)^2$

Q.18 The minimum value of ω_0 below which the ring will drop down is

[A] $\sqrt{\frac{g}{\mu(R-r)}}$ [B] $\sqrt{\frac{2g}{\mu(R-r)}}$ [C] $\sqrt{\frac{3g}{2\mu(R-r)}}$ [D] $\sqrt{\frac{g}{2\mu(R-r)}}$

END OF PART I : PHYSICS

Answers for the above questions

Q.17: Due to internal review, all candidates are awarded +3 marks.

Ans for Q.18: (A)